

### ■ B.3.8 Mathematical Functions

The mathematical functions such as `Log[x]` and `BesselJ[n, x]` which are built into *Mathematica* have a number of features in common.

- They carry the attribute `Listable`, so that they are automatically “threaded” over any lists that appear as arguments.
- They give exact results in terms of integers, rational numbers and algebraic expressions in special cases.
- Except for functions whose arguments are always integers, mathematical functions in *Mathematica* can be evaluated to any numerical precision, with any complex numbers as arguments. If a function is undefined for a particular set of arguments, it is returned in symbolic form in this case.
- When possible, symbolic derivatives and integrals of built-in mathematical functions are evaluated in terms of other built-in functions.